

E314 Final Design Report

Low-Power Embedded Light Sensor System

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1. System Executive

Summary

The goal of this project and the combination of all the labs was to design and implement a low-power embedded sensing system capable of periodically measuring ambient light and environmental conditions. The intended customer for this system is a user requiring a compact, battery-powered sensor for long-term deployment, such as environmental monitoring, logging, or educational applications.

Over the past semester, the system was developed one step at a time through a sequence of 10 total labs that addressed software development, sensor interfacing, data logging, hardware design, enclosure design, and power & cost analysis. The final design consists of an STM32L432 microcontroller, a TSL237 light sensor, on-chip temperature and voltage sensing, non-volatile data storage using internal flash memory, and a custom printed circuit board housed inside a sealed enclosure with a clear acrylic lid. Emphasis was placed on low-power operation, modular software design, and manufacturability.

2. System Description

The completed system is an embedded light-sensing platform designed to periodically collect data and store it for later retrieval. The hardware consists of a custom two-layer PCB populated with an STM32L432 microcontroller, a TSL237 light-to-frequency sensor, a programming/debug interface, and supporting passive components. The system is powered by a CR2032 coin-cell battery.

From a software perspective, the system provides a command-line interface over UART that allows users to interact with the device, configure parameters, view sensor readings, retrieve logged data, and erase or reset stored information. The firmware is interrupt-driven and makes use of timers, ADC peripherals, RTC

functionality, and internal flash memory.

Mechanically, the PCB is enclosed within a custom 3D-printed enclosure base and sealed with a laser-cut clear acrylic lid. The enclosure physically supports the PCB, protects it from environmental exposure, and allows light to reach the sensor.

3. Software Design

3.1 High-Level Software

Architecture

The system software was developed incrementally throughout Labs 2-6. At a high level, the firmware consists of the following components:

UART-based command-line interface

Interrupt-driven serial input buffering

Real-time clock (RTC) and alarm handling

Sensor acquisition (light, temperature, battery voltage)

Flash data logging and retrieval

Power-aware execution flow

3.2 Command-Line Interface and UART

Initial system bring-up was performed using STM32CubeIDE and STM32CubeMX to generate a template project. UART communication was configured to allow basic command entry for controlling the on-board LED. In Lab 3, the UART interface was extended to use an interrupt-driven receiver with a circular buffer, allowing non-blocking command input.

A flexible command-execution framework was implemented using function pointers and string parsing. Commands such as help, lon, lof, and test were implemented to demonstrate extensibility. This approach enabled later labs to easily add new commands without restructuring the core software.

3.3 Timekeeping and Scheduling

The real-time clock (RTC) was configured using CubeMX and used to maintain system date and time. Commands were introduced to set and display the current time and date, and the command prompt was updated dynamically to reflect the current timestamp.

An RTC alarm interrupt was configured to trigger periodically. Rather than performing actions directly inside the interrupt handler, a flag-based approach was used to signal the main loop, ensuring reliable and maintainable interrupt behavior.

3.4 Sensor Acquisition

Multiple sensors were integrated into the system. The TSL237 light sensor was interfaced using timer input capture to measure output frequency as a function of ambient light. Additional commands were created to read the microcontroller's internal temperature sensor and to measure battery voltage using the ADC and internal reference voltage.

Each sensor reading could be accessed via the command-line interface, enabling testing, validation, and demonstration of functionality.

3.5 Flash Data Logging

To support long-term operation, data storage in internal flash memory was implemented. Structured records were defined for both sensor data and event log messages. Date and time information from the RTC was packed into 32-bit fields to conserve memory.

Commands were provided to sample sensor data, retrieve stored records, and erase flash memory when needed. This functionality enables the system to operate autonomously and later provide historical data without continuous communication.

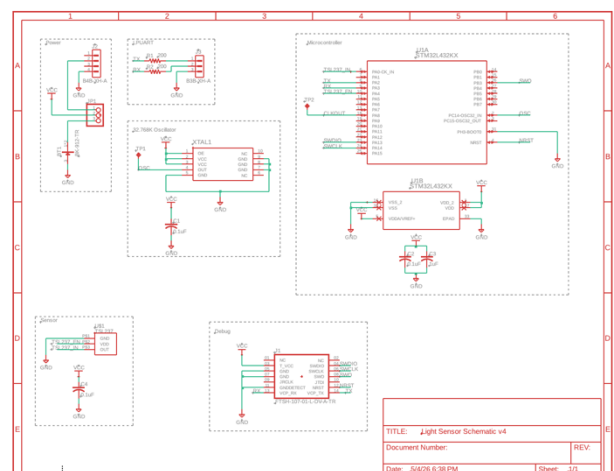
3.6 Power Management and Modeling

Power consumption analysis was performed using the STM32CubeMX Power Consumption Calculator. Various combinations of running and low-power modes were evaluated, and external sensor current was manually included based on the TSL237 datasheet.

Results showed that operating in low-power run mode combined with Stop or Standby modes significantly improved battery life. These findings informed recommended operating strategies for long-term deployment.

4. Hardware Design

4.1 Hardware Block Diagram



4.2 Schematic Design

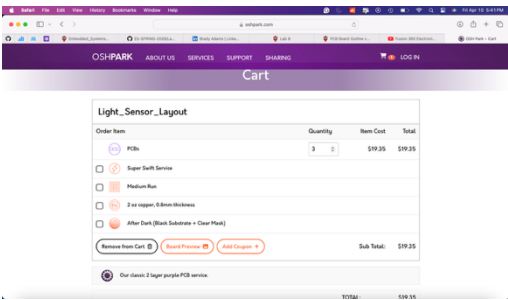
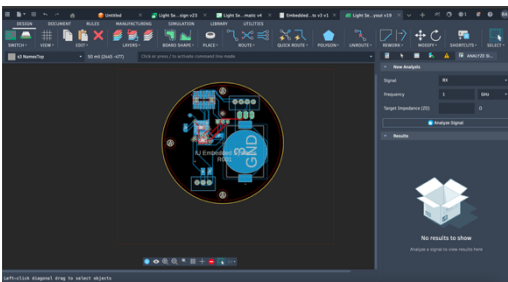
The hardware schematic was created using Autodesk Fusion 360 Electronics. Major components include the microcontroller, light sensor, debug/programming interface, clock components, power connections, and decoupling capacitors. Electrical rules checks were performed to ensure schematic correctness.

A bill of materials (BOM) was generated listing part numbers, suppliers, and estimated costs. Alternative microcontrollers were analyzed as part of the design evaluation, considering factors such as power consumption, peripherals, and availability. The STM32L432 was

selected due to its low-power capabilities and peripheral set.

4.3 PCB Layout

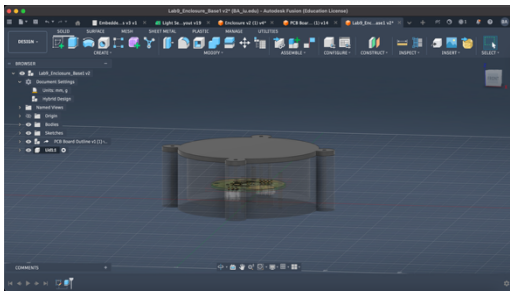
The PCB layout was completed in Fusion 360 Electronics following Osh Park two-layer design rules. Component placement prioritized signal integrity, power distribution, and compatibility with the mechanical enclosure. A ground plane was implemented to reduce noise, and silk-screen labeling was cleaned to improve manufacturability.



5. Enclosure Design

The enclosure consists of a 3D-printed base and a laser-cut clear acrylic lid. The base includes integrated standoffoffs to support the PCB and prevent movement during operation. An O-ring groove helps provide environmental sealing, while an access opening allows connection to the communication interface.

The clear acrylic lid ensures that sufficient light reaches the TSL237 sensor while protecting the electronics. Based on the design, a NEMA 3-equivalent enclosure rating is appropriate, as it provides protection against dust and light moisture, which is suitable for indoor or lightly protected outdoor deployment.



6. System Cost Modeling

Prototype cost estimation was performed for system quantities of 5, 20, 50, and 100 units. PCB fabrication and assembly costs were obtained using MacroFab, 3D enclosure printing costs using Shapeways, and laser-cut lid costs using Sculpteo.

The analysis showed that PCB costs decrease significantly with increased quantity due to manufacturing efficiencies, while enclosure costs scale more linearly. Recommendations were made to reduce cost for higher volumes, such as transitioning away from 3D printing.

7. Safety Considerations

Potential failure modes include battery depletion, corrupted flash memory, and sensor malfunction. Improper disposal of lithium coin-cell batteries presents environmental and safety risks. The system does not pose significant physical hazards during normal operation, but proper handling and disposal procedures should be followed.

8. Ethical Considerations

Although the system does not directly implicate ethical concerns such as privacy or safety-critical operation, responsible engineering practices were followed throughout the design. The IEEE Code of Ethics emphasizes honesty, reliability, and public welfare. Designing for safe operation, accurate measurement, and clear documentation aligns with these principles.

My background in programming and interest in programming influenced the overall structure and robustness of the firmware and system design.

9. Future Work

Future improvements could potentially include wireless communication, more advanced power management techniques, external non-volatile memory, and environmental sealing improvements. Additional sensors could also be integrated to expand the system's application.

10. Team

Throughout the semester, every project/lab was completed by our 2-person team. Both of us have contributed significantly throughout the semester, with responsibilities divided to leverage individual strengths in software development, hardware testing, and system integration.

10.1 Brady Adams

I am a third-year undergraduate student studying Computer Engineering with a minor in Intelligent Systems Engineering. I have a strong interest in low-level programming in C and the integration of AI inside game development environments.

Throughout this course, I was primarily responsible for the software development and system architecture of every project. This included configuring projects using STM32CubeMX and STM32CubeIDE, implementing the UART-based command-line interface, writing interrupt-driven firmware, and developing support for sensors, RTC functionality, flash memory logging, and power-aware operation. I also completed the majority of the Fusion 360 Electronics work, which included schematic design, PCB layout, and iteration. In addition, I also contributed heavily to system-level integration, debugging, and analysis tasks such as power estimation and prototype cost modeling.

10.2 Jaiden Medina

I am a third-year undergraduate student studying Computer Engineering with minors in Intelligent Systems Engineering & Mathematics. I have a strong interest in systems that bridge hardware and software, particularly in practical, real-world embedded applications. Previous coursework and projects involving embedded computing and performance-constrained systems helped shape my contributions to this project.

During the semester, I focused primarily on hardware testing, physical system validation, and project organization. This included building and testing the system on physical hardware and verifying sensor functionality. I also took responsibility for managing the GitHub repositories, ensuring that code commits were properly organized and documented, and maintaining clear and consistent README.md files to describe system functionality and usage.

My hands-on testing and validation efforts were critical for confirming that the software and hardware worked together reliably, and for identifying issues that required iteration or refinement.

10.3 Team Collaboration

Both of us collaborated closely on system design, debugging, and verification throughout the semester. While responsibilities were divided, key choices such as power management strategies, enclosure constraints, and system functionality were agreed upon jointly. This division allowed us to efficiently develop a complete embedded system while maintaining strong integration between hardware and software components.

11. References

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